

Joseph Itopa, Abubakar



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- Autonomous critical thinker, with broad analytical skills with around 6 years of experience in conceptualizing, managing and applying problem solving skills to data projects.
- Proficient engineer, with practice in data collection, processing, analysis, modeling, deployment and communication through end-to-end data science projects done in large (50+ members) and international collaborations.
- Versatile analyst, with broad experience in implementing and delivering data-driven solutions to real world problems such as disaster response, environmental protection and object detection in satellite imagery, in projects carried out in collaboration with private and humanitarian organizations.

• WORK EXPERIENCE

Senior M.L. Engineer – Vendease Inc. [Location: Lagos] | 01/2023 – 03/2024 (Job Type: Full time)

Responsibilities:

- Developed and deployed LLM bot using Retrieval Augmented Retrieval, FastApi, and Lanchain, for internal usage(company) on historical insights retrieval.
- Developed Chatbot for customer interaction with insights using LangChain, FastApi, and MLFlow.
- Research, development, tracking, and serving of demand forecast pipeline for demand planning.
- Development of credit risk machine learning models for the finance department.
- Lead a team to develop monthly market insight reports for customers in Nigeria, and Ghana.

Lead Data Science Engineer – Kobo360 [Location: Lagos] | 05/2021 – 12/2022 (Job Type: Full time)

Responsibilities:

- Implement and deployed LLM bot for real-time trip incidence enquiry using LangChain, and FastApi.
- Development of an ML pipeline for extraction of trip records, and predicting trip cost estimation.
- Developed and deployed automated ML pipeline on AWS Lambda for trip price estimators; for Nigeria, Kenya, and Ghana.
- Experimentation, training, and deployment of statistical model for trip incidence prediction in Nigeria.

Product Owner/Manager – Omdena [Location: Remote] | 06/2021 – 10/2021 (Job Type: Volunteering)

Responsibilities:

- Manage over 30 Artificial Engineers from across the globe, to develop AI solution.
- Organized weekly scrum meeting with AI engineers and present weekly report to Stakeholders.
- Lecturing over a 100 students on MLOps for AI engineers and data scientist.

Machine Learning Engineer – Omdena [Location: Remote] | 03/2020 – 03/2021 (Job Type: Volunteering)

Responsibilities:

- Trained a pre-trained resnet18|34|50 CNN models with enhanced images.
- Preprocessing satellite images using Histogram equalization techniques.
- Build both automated machine learning models: IBM Watson; and non-automated machine learning models: Gradientboost, Random Forest, Support Vector, Prophet, and Linear-learner on Sagemaker.
- Scraping and wrangling of disaster response data.
- Explored time series data - global disaster response data, and student debt data.

Data Analyst | Technical Manager – Inter-Trade Ltd | Nigeria] | 04/2018 – 10/2020 (Job Type: Full-time)

Responsibilities:

- Analyze location based data for project locations different regions of Nigeria.
- Generated insights from the financial data for UNICEF/WASH reports.
- Selecting and supervising teams for clients' energy demands evaluation and presenting possible renewable energy solutions.
- Trained teams in Uganda, and Sudan on management and utilization of a portal for Internet of Things products: Solar Cold-Chain Equipment, and IoT platform.

• VOLUNTEERING EXPERIENCE

Role	Program/Event	Organization
Coach/Mentor	Startup Incubator	TechQuartier
Data Science Tutor	Workshop	Women-Who-Code
Mentor	Workshop	International Conference for Machine Learning (Google)
Mentor	Workshop	DataNative
Python Tutor	Workshop	DjangoGirls

• EDUCATION

Certificate	Date	Organization
MSc Big Data Technologies	July, 2024	University of East London
Google Professional Data Engineer	February, 2024	Google Inc
Machine Learning Scientist Track	December, 2023	Datacamp
Machine Learning (Health and Finance Track)	August, 2022	Oxford ML Summer School
AWS Machine Learning Specialty	July, 2021	CloudAcademy
Data Engineering Track	October, 2022	Datacamp
ETL in Python	August, 2022	Datacamp
Building Data Eng'g Pipelines in Python	September, 2022	Datacamp
Google Project Management Certificate	April, 2022	Cousera
Advance NLP with spaCy	February, 2022	Datacamp
Credit Risk Modelling in Python	October, 2021	Datacamp
Bachelor of Engineering in Telecommunication Engineering	December, 2015	Fed. Uni. Tech. Minna

• SKILLS

TECHNICAL SKILLS

Programming Languages and Libraries

Python | PostgreSQL | MySQL | Weight & Bias | Pyspark | MLFlow | LangChain | Keras | Sklearn | OpenAI

Tools

AWS Sagemaker | Docker | AWS s3 | Trello | Jira | Confluence | Airflow | Prefect | Github | AWS Boto3 | Google PubSub | Google Vertex AI | Google Dataflow | Google Cloud Storage | BigQuery | PowerBI | Kafka | Hadoop

SOFT SKILLS

Technical Writing | Project Scoping and Planning | Critical and Analytical Thinking | Time Management | Effective Communication

• OTHER PROJECTS

- LLM bot to query insights public on public health data for Nigeria, using FastApi, Lanchain, OpenAI.
- Ochestrated ChatGPT Tweet Sentiment Analysis using Hadoop, Pyspark, and Airflow.
- [Predicting Bed Utilization Rate during Pandemic](#)
- Developed and deployed credit risk analysis and modelling for a UK startup on Google Cloud.
- [Increasing Clean Energy Access in Africa through Predictive Modeling](#)
- [Using Satellite Imagery to Detect and Assess the Damage of Armyworms in Farming](#)
- [Preventing Malaria Infections through Topography and Satellite Image Analysis](#)
- [AI for Disaster Response: Improving Emergency Management during Cyclones](#)
- Evaluation of ROC and F1-score for Quality Assessment of CNN image classifier.
- Migration of Customer transaction data from MySQL & Postgres DB to Bigquery & Snowflake using Airbyte.
- [Data Modelling for Public Health in Nigeria using Kimball Modelling Technique\(Star Schema\)](#).
- Developed Orchestrated Batch Pipeline with Prefect for Vehicle Accident Analysis.
- Developed Orchestrated Batch Pipeline with Prefect for clustering of Public Health data.
- Implementation of Orchestrated Batch Pipeline for Police Incidence Record using PySpark, Airflow, & Bigquery.